Aryan Dayal

(609) 285-7796 | dayal4@purdue.edu | www.linkedin.com/in/aryan-dayal

EDUCATION

Purdue University, School of Mechanical Engineering — **West Lafayette, IN**
Bachelor of Science in Engineering — **Expected Graduation: May 2027**

- **Awards:** Best Research Poster (Purdue SURF 2025), Dean's List, Semester Honours — **GPA**: 3.8/4.0

RELEVANT SKILLS

- **Micro/Nano Fabrication:** Cleanroom Certified, Photolithography, Thin-Film Deposition, SEM, Optical Microscopy
- **Programming & CAD:** Python, MATLAB, Pandas, NumPy, Machine Learning, Siemens NX, Simulink, GD&T, LLMs
- **Shop/Prototyping:** CNC Machining, Design of Experiments (DOE), Process Optimization, Yield Improvement, 3D printing

PROFESSIONAL EXPERIENCE

Purdue University – School of Mechanical Engineering — **West Lafayette, IN**
Undergraduate Researcher (Nanoscale Energy Transport & Conversion Lab) — **January 2026 – Present**

- Developed and validated **automated simulation workflows** (LAMMPS, Phonopy) for thermal transport analysis in crystalline materials, processing **10^3–10^4 atomic configurations** across large datasets and complex parameter spaces
- Built **Python-based automation pipelines** for **1000+** parameter sweeps, reducing manual post-processing time by **~70%** while leveraging Large-Language Models (LLMs) for workflow acceleration and data processing, improving data throughput
- Performed **large-scale data analysis, root-cause analysis,** and **troubleshooting** of simulation outputs, identifying convergence failures, validating physical accuracy, and supporting **data-driven process optimization**

Purdue University – School of Materials Engineering — **West Lafayette, IN**
Undergraduate Researcher (NASA-Funded | Martinez Soft Materials Group) — **May 2025 – Present**

- Fabricated and characterized **3 µm silica monolayers** and **10 nm Pt thin films** using **sputter deposition**, **SEM**, and **optical microscopy**, performing calibration, metrology, and troubleshooting to ensure fabrication repeatability
- Developed electric-field-assisted assembly methodologies for amphiphilic Janus particles (Si-Pt), conducting **25+ experimental trials** and establishing a **phase diagram** correlating electric-field conditions with microstructure formation
- Designed and executed **Design of Experiments (DOE)** studies (**50+ test conditions)** utilizing **Python-based** data analysis to evaluate particle assembly, identify failure modes, optimize process parameters, and improve fabrication consistency

Purdue University – School of Mechanical Engineering — **West Lafayette, IN**
Undergraduate Research Lead (Manufacturing Lab) — **January 2025 – December 2025**

- Designed and optimized **Gaussian-to-top-hat beam shaping systems** for 800 nm optics, translating simulation outputs into fabrication-ready geometries to achieve **≤5% intensity non-uniformity** through iterative design validation and analysis
- Applied D**esign of Experiments (DOE)** and response optimization methods across **20+** design iterations, improving transmission efficiency to **≥85%** while satisfying manufacturability constraints and system performance targets
- Fabricated and characterized micro-scale devices using **two-photon lithography (TPP), SEM,** and **sputter deposition,** performing metrology, defect analysis, and process refinement to improve structural fidelity and fabrication reliability

LEADERSHIP & INVOLVEMENT

Purdue University – School of Mechanical Engineering — **West Lafayette, IN**
Tutor — **January 2026 – June 2026**

- Provided **one-on-one tutoring** in core Mechanical Engineering courses (**Mechanics of Materials, Systems, Measurements, and Controls**), supporting **30+ students** with problem-solving, exam preparation, and technical concept reinforcement
- Guided students through **free-body diagrams**, **equilibrium analysis**, and **energy balances** using **first-principles reasoning**

Purdue Department of Physics and Astronomy — **West Lafayette, IN**
Teaching Assistant: Lab — **August 2024 – December 2024**

- Guided **120+ students** through VPython modeling and labs, emphasizing **first-principles reasoning**, **measurement procedures**, and **Six-Sigma-style** data reduction to produce presentation-quality results
- Supervised instrumentation labs using **oscilloscopes**, **multimeters**, **Hall probes**, and **sensor-based** measurement systems; verified experimental setups and performed calibration checks to ensure data repeatability and reliability

PROJECTS

Monolayer Deposition System: Designed and fabricated a motor driven deposition platform for large-area self-assembled particle monolayers, integrating custom mechanical components and process optimization techniques to improve deposition uniformity
Automated Microscopy Image Analysis: Developed Python-based image processing workflows to quantify particle assembly behavior from microscopy images, enabling automated extraction of process metrics and experimental performance trends